Chapter test

1 Introduction to biology

Name: _____ ( ) Class: _____ Date: _____

Total marks: 25 marks
Time allowed: 25 minutes
Answer ALL questions.

Score: / 25

A. Concept checking (10 marks, 1 mark each)

1. Which of the following processes is ***not*** an example of excretion?

   A. breathing out carbon dioxide from the lungs
   B. passing undigested food from the anus
   C. removing urea through urination
   D. excess oxygen formed during photosynthesis ☐

2. The body temperature of humans is about 37 ^{o}C. Which of the following processes produce(s) heat to maintain a constant body temperature?

   (1) respiration
   (2) movement
   (3) growth

   A. (1) only
   B. (2) only
   C. (1) and (2) only
   D. (1) and (3) only ☐

3. Which of the following is ***not*** evidence that plants are living organisms?

   A. Plants are green in colour.
   B. Plants carry out respiration all the time.
   C. Plants excrete by storing the waste products in leaves which then fall off in dry seasons.
   D. The roots of plants grow towards water. ☐

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4. Which of the following correctly matches the branch of biology with its scope?

	Branches of biology	***Study of***
A.	cytology	structure of body systems
B.	physiology	how organisms function
C.	ecology	classification of organisms
D.	anatomy	chemical processes in organisms

5. Which of the following areas involve the application of biological knowledge?

(1) preservation of food
(2) production of medicines
(3) conservation of the environment

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

6. What is a hypothesis?

A. a method to examine the idea proposed by scientists
B. a prediction of the result of an experiment
C. a proposed explanation of observations
D. a method to analyse experimental data

Directions: Questions 7 to 10 refer to the following investigation carried out by a student. The student wondered if drinking milk could improve health. He invited his classmates to join his investigation. The students were divided into two groups:

Group 1: 15 students drink a glass of milk before breakfast every day for a month.

Group 2: 15 students drink a glass of water before breakfast every day for a month.

During the month, the health condition of the students was recorded and the results were summarized below.

	Group 1	**Group 2**
Number of students who had caught a cold	2	2
Number of students who had had diarrhea	4	6

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7. Which of the following is the independent variable of the investigation?

A. whether the students eat breakfast or not
B. whether the students drink milk or not
C. whether the students catch a cold or not
D. whether the students have diarrhea or not ☐

★ 8. Which of the following assumption(s) should have been made so that a valid conclusion could be drawn?

(1) All students are in good health condition at the beginning of the investigation.
(2) The type and amount of milk drunk by the students are the same.
(3) The consumption of milk is the only factor that affects the students' health.

A. (3) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

★ 9. The reliability of the investigation can be improved by

(1) involving more students in each group
(2) lengthening the duration of experiment
(3) asking Group 2 students to drink fruit juice instead of water

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

★ 10. Based on the results, what conclusion can be drawn?

A. Drinking milk can improve health.
B. Drinking milk cannot improve health.
C. Drinking milk can prevent diarrhea.
D. The results did not provide solid evidence for drawing a conclusion. ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	1.1	1.1	1.1	1.2	1.3	1.3	1.3	1.3	1.3	1.3

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B. Problem solving (6 marks)

11. Mary's mother put a cucumber into the refrigerator. After ten days, Mary found that the cucumber became soft. She wondered why this happen and carried out a scientific investigation on this. Identify the steps and fill in the blanks with suitable words from the list below. (6 marks)

analysis	ask a question	assumption	conclusion
experiment	hypothesis	observation	prediction

(a) Mary noticed that the cucumber became soft after putting it into the refrigerator for ten days. _____

(b) Mary wondered why the cucumber became soft. _____

(c) She suggested that it might be because the cucumber lost water to the fridge. _____

(d) She put two rigid and fresh cucumbers into the fridge. One of them was wrapped with kitchen wrap while the other did not. _____

(e) Mary thought that the kitchen wrap can prevent the cucumber from losing water. _____

(f) Mary guessed that the cucumber wrapped with kitchen wrap would remain rigid while the cucumber without wrapping any kitchen wrap would become soft. _____

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C. Exam challenge (9 marks)

★★ 12. Read the passage below and answer the questions that follow.

In the late 1920s, Alexander Fleming was searching for an effective drug for treating bacterial diseases. He studied the development of a type of bacteria called *Staphylococcus*. He cultured them on agar plates. Normally Fleming covered his bacterial colonies on the plate with a lid to prevent contamination. But one day, he accidentally left a dish uncovered near an open window.

After spending a month on holiday, Fleming returned to his laboratory and found that a mould was growing on the agar. He noticed that the bacteria could not grow close to the mould. Fleming identified the mould as *Penicillium notatum.* He suggested that the mould produced a substance which killed the bacteria and named the substance as penicillin. This was the first antibiotic discovered. Fleming believed that penicillin had the potential to become a 'wonder drug', but he was unable to isolate it in large quantities and keep it active long enough to be used in humans.

Ten years later, Howard Florey and Ernst Chain found ways to extract large quantities of active penicillin in the mould.

(a) With reference to the above passage, identify ***two*** characteristics of *Staphylococcus* showing that it is a living organism. (2 marks)

(b) What did Fleming observe after he returned to the laboratory? (2 marks)

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(c) Based on the observation, what hypothesis did Fleming make? (1 mark)

(d) Other than the ability to kill bacteria, suggest another property of penicillin which makes it a useful drug. (1 mark)

(e) In the experiment that Fleming discovered penicillin, there were some experimental practices that should be avoided when studying bacteria. State ***two*** malpractices of Fleming and explain why those practices were dangerous. (3 marks)

(Total: 9 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	B	2.	C	3.	A	4.	B	5.	D
6.	C	7.	B	8.	D	9.	A	10.	D

(1 mark each)

B. Problem solving (6 marks)

11. (a) observation (1)
    (b) ask a question (1)
    (c) hypothesis (1)
    (d) experiment (1)
    (e) assumption (1)
    (f) prediction (1)

(Total: 6 marks)

C. Exam challenge (9 marks)

12. (a) They grow in number. (1)
    They obtain nutrients from the agar plate. (1)
    (b) He found that a mould was growing on the agar. (1)
    He noticed that the bacteria could not grow close to the mould. (1)
    (c) The mould produced a substance which killed the bacteria. (1)
    (d) It is non-toxic to humans. /
    It remains effective after extraction. /
    It can be easily isolated (or purified). (1)
    (e) He left the plates uncovered. (1)
    He left the plates near an open window. (1)
    The bacteria might spread to the environment through the air. (1)

(Total: 9 marks)

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